Ecology & Ecosystem.

1. What is Ecology?

Ans Ecology is the branch of biology that studies the interactions between living organisms (plants, animals, microbes) and their physical environment (air, water, soil, sunlight, temperature, etc). It examines how organisms depend on each other and how they are influenced by both biotic (living) and abiotic (non-living) components of the environment.

. The term "Ecology" was first introduced by Ernst Haeckel, a German zoologist, in 1866. He combined two Greek words: "Oikos" meaning 'house' or 'place to live' and "Logos" meaning 'study of'. So, ecology literally means 'study of the home of living things.'

Main Aim of Ecology:

. To understand how organisms are distributed across the earth.

. To study the abundance (population size) of organisms.

. To investigate the interactions between organisms and their environment.

. To understand the distribution of biotic and abiotic factors and their mutual relationships.

Ex- studying how tigers interact with their prey, water availability, and vegetation in a forest ecosystem.

2. What is an Ecosystem?

An ecosystem is a functional unit of nature where living organisms (plants, animals, microbes) interact with each other and with the non-living environment (climate, soil, water, minerals) to sustain life through nutrient cycling and energy flow.

- Introduced by: A.G. Tansley in 1935.
- Structure: Every ecosystem has biotic components (living) and abiotic components (non-living). Both work together to maintain ecological balance.

Types:

- Natural eco-systems → Forests, grasslands, deserts, ponds, lakes, rivers, oceans.
- Man-modified ecosystems → Agricultural fields, urban systems, industrial areas.

Ex - A pond ecosystem includes fish, algae, aquatic plants, bacteria (biotic) interacting with water, sunlight, minerals and air (abiotic).

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3. What are the types of Ecosystems Services?

Ans Ecosystem services are the benefits humans obtain from ecosystems. They are vital for survival and quality of life. They are classified into four types:

1. Provisioning services:

- Tangible products obtained from ecosystems

Ex → Food (crops, fish, fruits, meat), fresh water, medicines from plants, timber, biofuel.

2. Regulating services:

- Services that maintain environmental balance

Ex → Air purification by forests, control of greenhouse gases, soil erosion prevention, pollination by insects, pest control by birds, flood regulations by wetlands.

3. Supporting services.

- Services that form the base for all other services.

Ex → Nutrient cycling, soil formation, Oxygen production, providing habitat for species, biodiversity maintenance.

4. cultural services:

. Non-material benefits human get from ecosystems.

Ex → Rec Recreation (nature tourism), spiritual inspiration (sacred forests), cultural identity, ~~aesthe~~ aesthetic values.

9. What are the two main components of an Ecosystem?

Ans An ecosystem has two major components.

A) Biotic components (Living):

- producers (plants, algae), consumers (animals) decomposers (bacteria, ~~fun~~ fungi).

B) . Depend on abiotic factors for survival.

B) Abiotic components (Non-Living):

. physical factors: climate, light, temperature, water.

. chemical factors: Minerals, gases, nutrients, soil types.

Interactions: Biotic components use abiotic resources to maintain the flow of energy in the ecosystem.

5. What are the Biotic and Abiotic Factors?

Ans

- Abiotic Factors : Non-living factors in the ecosystem such as sunlight, air, water, soil, minerals, temperature, pH. They determine what organisms can live in an area.

- Biotic Factors: Living components such as plants, animals, fungi, bacteria.

They can be:

1. Autotrophs (Producers) : Produce their own food (e.g. plants).
2. Heterotrophs (consumers) : Depend on other organisms for food (e.g. animals).

6. Who are Producers?

Ans

- Producers are autotrophic organisms (mostly green plants and algae) that make their own food by photosynthesis using sunlight, water and carbon dioxide.

- They form the first trophic level and are the main source of energy in an ecosystem.

Ex → Algae in ponds, grasses in fields, trees in forests.

By-product → Oxygen.

7. Who are consumers?

Ans Consumers are heterotrophic organisms that depend on producers or other consumers for food.

Types -

1. Primary consumers (Herbivores): Feed directly on plants (ex, deer, cow, rabbit).

2. Secondary consumers (carnivores / omnivores): Eat herbivores (ex, snakes, dogs, cats).

3. Tertiary consumers: Eat secondary consumers; usually top carnivores (ex, tiger, lion).

4. Quarternary consumers: Feed on tertiary consumers (ex, Eagle eating snake that ate a frog).

8. What are decomposers (Reducers ~~Reducers~~ and Transformers)?

Ans Decomposers are organisms (mainly fungi and bacteria) that breakdown dead plant and animal matter into simpler substances.

- Essential for recycling nutrients back into the soil.

Ex → Fungi, bacteria, actinomycetes.

9. What are types of Ecosystem?

Ans Ecosystems can be broadly classified into two main types based on their location and characteristics:

- Aquatic Ecosystems: These are ecosystems found in water bodies. Examples include oceans, seas, rivers, lakes, ponds, and wetlands.

  Aquatic ecosystems cover about two-thirds of the Earth's surface and support a wide variety of life forms like fish, algae, amphibians, and corals.

- Terrestrial Ecosystems: These are ecosystems found on land, such as forests, grasslands, deserts, tundra and agricultural lands.

  ~~such as~~ Terrestrial ecosystems are greatly influenced by climate and vegetation types.

10. What is an Aquatic Ecosystem? Explain its types.

Ans An aquatic ecosystem includes all water bodies like oceans, rivers, lakes, ponds, and wetlands. These ecosystems provide habitats for diverse plants and animals adapted to living in water.

Aquatic ecosystem is divided into:

- Freshwater ecosystem - Rivers, lake, ponds, and wetlands where the salt content is very low. These cover about 3% of the Earth's water, but only about 0.5% is easily

accisible freshwater.

* Types of freshwater ecosystems:

- Lentic: Flowing water ecosystems like rivers and streams.
- Lentic: still water ecosystems like ponds and lakes.
- Wetlands: Areas where soil remains saturated most of the time, such as swamps and marshes.

• Marine ecosystems - Includes oceans, seas, coral reefs, and estuaries with high salt content. These cover around 71% of the Earth's surface and support a vast diversity of life such as whales, sharks, dolphins, and coral species.

11. What is Terrestrial Ecosystem? Describe its types.

Ans. A terrestrial ecosystem is a land-based community where living organisms interact with soil, air, and climate. They are mainly influenced by temperature, rainfall and sunlight.

Types of terrestrial ecosystem include:

• Forest Ecosystem - Areas dominated by trees and shrubs; can be tropical rainforests, temperature deciduous forests, or coniferous forests. Forests provide oxygen, shelter, and food for many species.

- Grassland Ecosystem - Dominated by grasses and herbs; found in regions with moderate rainfall. ~~Examines~~ Examples include savannas and ~~praises~~ prairies.

- Tundra Ecosystem - Found in polar regions with cold temperatures and little vegetation. Mostly mosses, lichens, and small shrubs grow here.

- Desert Ecosystem - Dry areas with very little rainfall and sparse vegetation. Organisms are adapted to survive extreme heat and drought conditions.

12. What are the functions of an Ecosystem?

Ans Ecosystems perform several vital functions-

- Regulation of Life Processes: Maintain balance among plants, animals, and microbes to support survival.

- Support for Nutrient cycles: Breakdown of dead matter recycles nutrients necessary for plant growth.

- Energy flow: Transfer of energy from the sun, through producers, consumers and decomposers.

- Maintain Ecological balance: Keep stable relationships between organisms and their environment.
- Provide Habitat: Offer living space for millions of species including humans.

These function ensures the sustainability of life on Earth.

13. Explain Energy Flow in an Ecosystem:

Ans: The sun is the ultimate source of energy for most ecosystems.

- Producers (plants) use sunlight in photosynthesis to make chemical energy stored in food.
- This energy passes to primary consumers (herbivores) when they eat plants.
- Secondary consumers (carnivores) gain energy by eating herbivores, and tertiary consumers eat secondary consumers.
- Decomposers break down dead organic material, returning nutrients to the soil.
- Energy flow is unidirectional (one-way) and decreases at each step because some energy is lost as heat.

This flow of energy sustains life in the ecosystem.

14. What is an Food Chain? Give an example.

Ans A food chain shows a simple, linear sequence of organisms where each one is eaten by the next. It demonstrates how energy and nutrients flow through different trophic (feeding) levels.

Ex → Grass (producer) → Goat (Primary consumer) → Human (secondary consumer).

The grass makes food using sunlight. The goat eats the grass, and the human eats the goat.

15. What are the components of an food chain?

Ans A food chain has four main components:

- Sun: Primary source of energy.
- Producers: Autotrophs like green plants and algae that produce food by photosynthesis.
- Consumers: Organisms that depend on producers or other consumers for energy.
- Decomposers: Organisms like fungi and bacteria that breakdown dead matter and recycle nutrients.

16. Give examples of food chain.

Ans

- Aquatic Food Chain:

Phytoplankton (Producer) → Zooplankton (Primary consumer) → small fish (Secondary consumer) → Larger Fish (Tertiary consumer).

- Terrestrial Food Chain:

Grass (Producer) → Butterfly (Primary consumer) → Frog (Secondary consumer) → Snake (Tertiary consumer) → Hawk (Quarternary Consumer).

17. What is a food web? How is it different from a food chain?

Ans A food web is a complex network of several interconnected food chains showing multiple feeding relationships among organisms in an ecosystem.

Differences from Food chain:

Aspect	Food Chain	Food Web
Definition	Linear pathway of energy flow.	Network of interconnected food chains.
Structure	Simple, one path	Complex with many paths.
Disturbance	One break can disturb disrupt the whole chain	More stable, as many alternative paths.

~~Food~~ Food webs better show the complex feeding relationships in nature.

18. What is Detritus Food Chain? Give Examples of Detritivores.

Ans A detritus food chain starts with dead organic matter (detritus) instead of living plants. Organisms called ~~detrivo~~ detritivores ~~eats~~ feed on this dead matter. They help decompose it, recycling nutrients back to the soil.

Ex of detritivores → Earthworms, beetles, millipedes, woodlice.

- The detritus food chain is important for nutrient cycling and ecosystem health.

19. Explain the Y-shaped Energy Flow Model.

Ans The Y-shaped energy flow model describes how energy flows through two interconnected food chains simultaneously.

- Grazing food chain: Energy flows from prod-ucers to herbivores to carnivores.

- Detritus Food Chain: Energy flows from dead organic matter to detritivores and decomposers.

The Y-shape shows these two chains separated at the base (producers v/s detritus) but connected through consumers and decomposers.

20. What is Ecological Succession? Describe it's types.

Ans. Ecological succession is the natural, gradual process by which ecosystems change and develop over time, with one community replacing another until a stable community is formed.

- Primary succession: Happens in life less areas where no soil exists, like after volcanic ~~erow~~ eruptions or glacier retreats. Life begins with pioneer species like lichens, and mosses.

- Secondary succession: Happens in areas ~~like~~ where a previous community was ~~distri~~ disturbed but soil remains, like after fires or floods.

Succession leads to increased biodiversity and ecosystem complexity.

21. What is a 'sere' in Ecological succession?

Ans. A 'sere' is the full sequence of communiti-es or stages in succession from the first colonizers (pioneer species) to the final stable climax community.

- Primary sere: Succession beginning on new land without previous life. (ex. bare rock).

- Secondary Sere - Succession after disturbance in an existing ecosystem.

22. What is Population and Community Ecology?

Ans
- Population - Group of individuals of the same species living in a specific area.
- Community - Different populations of species living and interacting in one area.
- Community Ecology - studies interactions among species in communities, their structures, abundance, demographics and relationships.

23. What is an Ecological Pyramid?

Ans An ecological pyramid is a graphical representation of the number, energy or biomass of organisms at each trophic level in an ecosystem.

Types of ecological pyramid:

- Pyramid of Numbers - shows the count of individual organisms ~~that~~ at each trophic level.
- Pyramid of Biomass - shows the total living mass at each trophic level.
- Pyramid of Energy - shows the amount of energy

available at each level; always upright.

24. Difference b/w Pyramid of Numbers, Biomass and Energy.

Ans

- Pyramid of Numbers: Represents the number of organisms at each level. Can be upright or inverted depending on ecosystem.
- Pyramid of Biomass: Represents total mass of organisms at each level. Can be inverted in aquatic ecosystems.
- Pyramid of Energy: Represents energy content at each level, always upright since energy decreases upward.

25. Define and explain Ecotone?

Ans An ecotone is a transition area where two distinct ecosystems or biomes meet and integrate, like where a forest meets a grassland. It usually has high species diversity and unique characteristi-cs.

Ex → Mangrove forests are an ecotone b/w marine and terrestrial ecosystems.

26. What is Edge Effect and Edge species?

Ans Edge effects refers to changes in species populations and community structures at the boundary (ecotone) between two habitats. Often, species diversity and population density can be higher in this zone. ~~Edge species~~

Edge species are organisms that thrive in these areas.

27. Define Habitat and Ecological Niche.

Ans

Habitat: The physical place or environment where an organism lives.

Ecological Niche: The role of function of an organism within its habitat, including how it gets food, interacts, and survives.

No two species can have identical niches in the same habitat.

28. Vocabulary: Important terms in Ecology.

Term	Definition
Carnivore	→ Organism that feeds on other animals.
Consumer	→ Organism that must consume others for energy.
Decomposer	→ organism that breaks down dead matter and wastes.
Food chain	→ linear sequence showing energy flow between organisms.
Food Web	→ Network of interconnected food chains.
Herbivore	→ Animals that eats plants.

Term	Definition
Producer	→ Organisms that makes its own food (autotroph).
Autotroph	→ Produces its own food (usually plants and algae).
Heterotroph	→ Organism that depends on others for food.
Ecotone	→ Transition zone b/w two ecosystems.
Edge Effect	→ change in population or community structure at edges.
Habitat	→ Place where an organism lives.
Ecological Niche	→ Functional role of a species in its environment.
Ecological Succession	→ Gradual change in species composition over time.